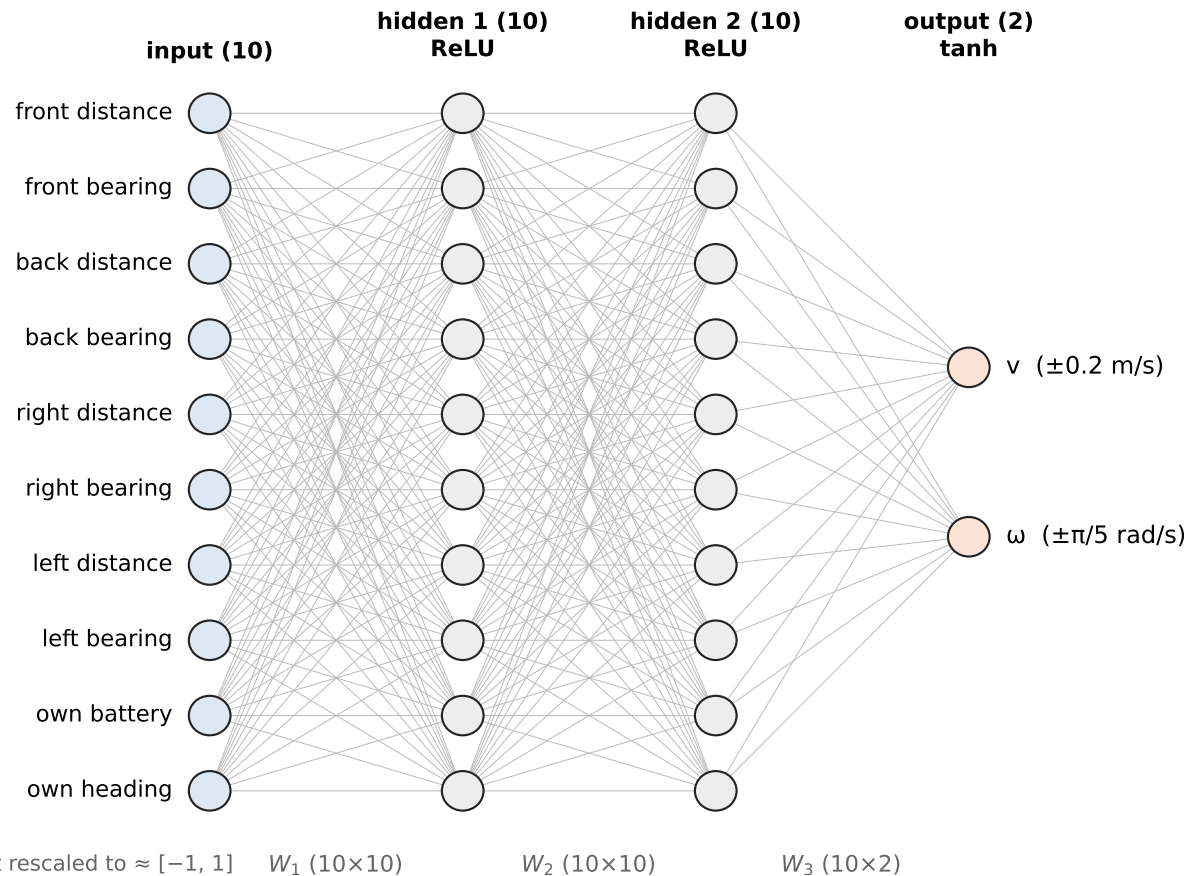




Every time step, every synapse

$$\Delta w_{ij} = \eta (A_{ij} n_i n_j + B_{ij} n_i + C_{ij} n_j + D_{ij})$$

n_i, n_j : pre-/post-synaptic activity
 $\eta = 0.1$; a layer is rescaled to
 $\max |w| = 1$ whenever it exceeds 1

What is evolved vs. learned

Evolved (CMA-ES): the genome — one
 (A, B, C, D) per synapse = 4×220
 = 880 numbers in $[-5, 5]$, shared by all robots.

Learned online (per robot): the weights
 W_1, W_2, W_3 — redrawn from $U[-1, 1]$
 at the start of every episode.